第二十七讲

静不定问题分析

作业：14-1,3(a),6,10(d)；15-14.

第十四章　静不定问题分析

14.1 静不定问题

14.2 用力法分析静不定问题

14.3 对称与反对称问题分析

14.4 平面刚架空间受力分析

~~14.5 连续梁与三弯矩方程~~

第十四章 静不定问题分析

14.1 静不定问题

1、多余约束与多余约束反力

2、静不定度的判断

1、多余约束与多余约束反力

静定问题　未知力数＝有效平衡方程数 → 可求内力

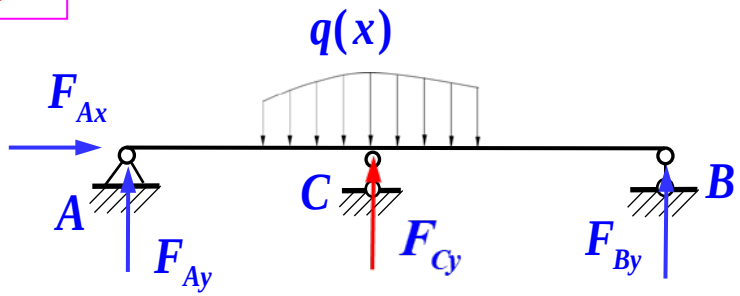
静不定问题　(Statically Indeterminate Problems)
(超静定问题)

未知力数 > 有效平衡方程数 → 内力不可求

二者之差 —— 静不定度

多余约束力 ← 多余约束

- 与静定问题的根本区别
- 求解的关键



2、静不定度的判断

1）平面桁架

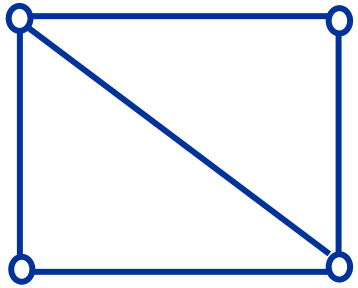
节点自由度：2； 杆件对节点的约束：1

A、无支座桁架

内力静不定度： $m - 2n + 3$； m：杆件数，n：节点数。

$4 - 2\times4 + 3 = -1$

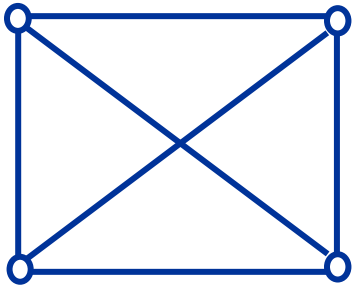
几何可变



$5 - 2\times4 + 3 = 0$

静定

或瞬时可变



$6 - 2\times4 + 3 = 1$

一度静不定

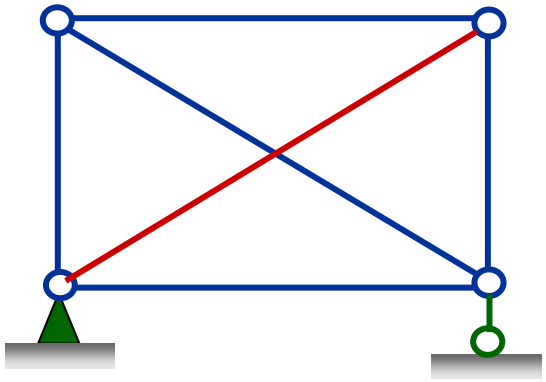
2、静不定度的判断

1）平面桁架

B、有支座桁架

内力静不定度： **m - 2n + k ；**

m：杆件数，n：节点数，k：支座约束数。



1）静不定度 = 5 +3 −2×4=0

2）静不定度 = 6 +3 −2×4=1

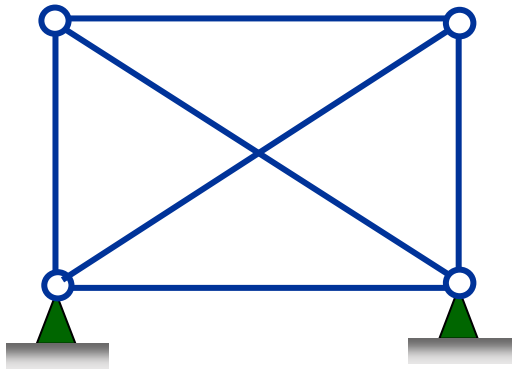
内力静不定(一度)

外力静定

2、静不定度的判断

1）平面桁架

B、有支座桁架



静不定度 ＝ 6－2×4＋4＝2

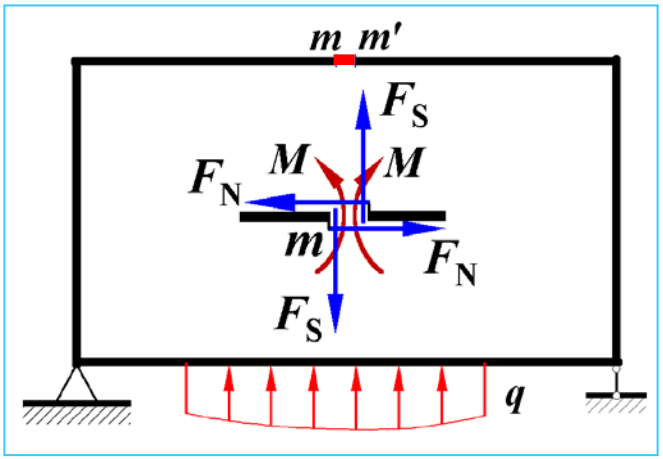
二度静不定

多余约束：

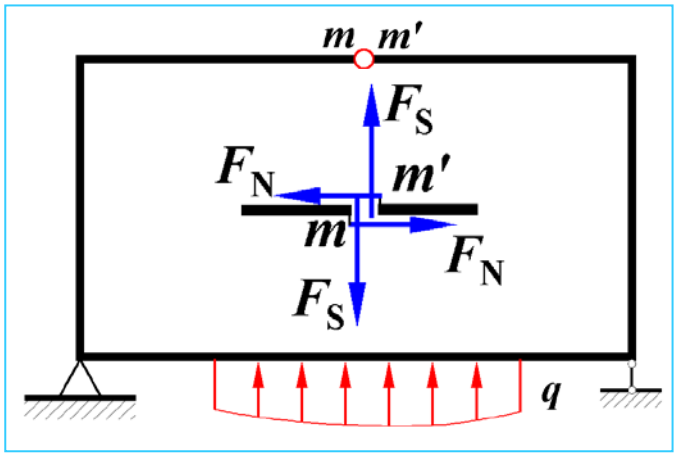
内部、外部各一个

2、静不定度的判断

2）平面刚架和曲杆



多余约束　3

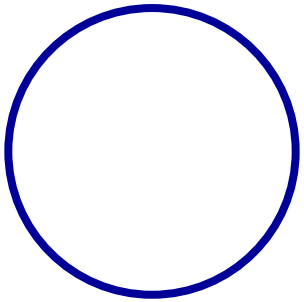


多余约束　2

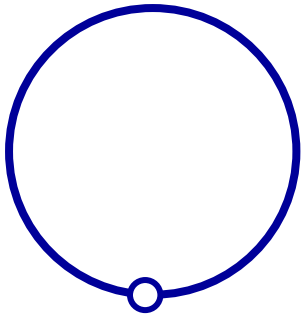
轴线为单闭合曲线的平面刚架或平面曲杆、且仅在轴线平面内承受外力时，为 3 度内力静不定问题

2、静不定度的判断

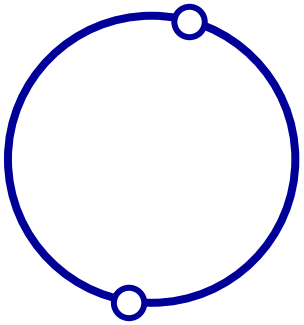
2）平面刚架和曲杆



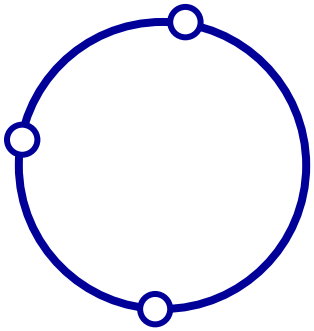
三度内力静不定



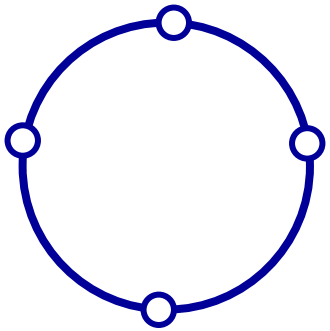
两度内力静不定



一度内力静不定



内力静定



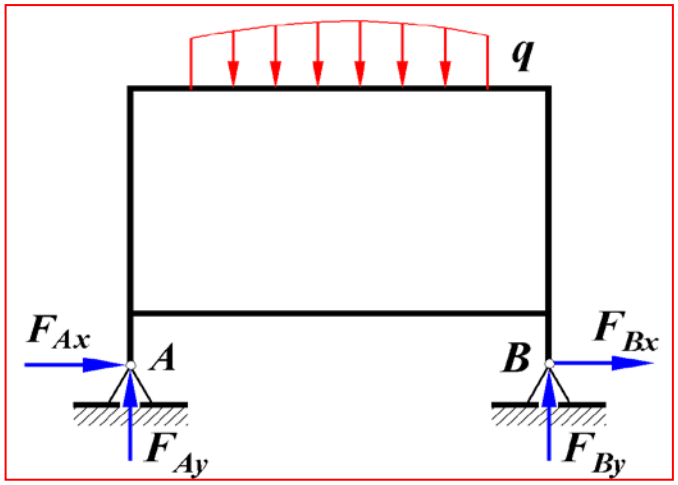
几何可变

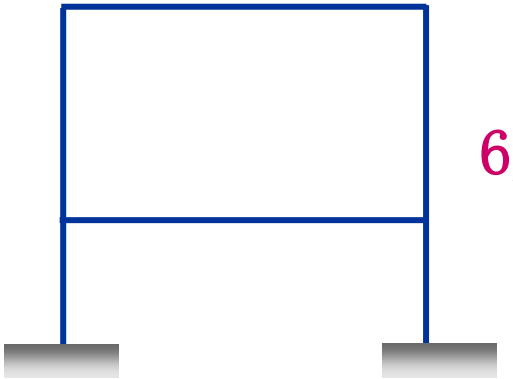
2、静不定度的判断

2）平面刚架和曲杆

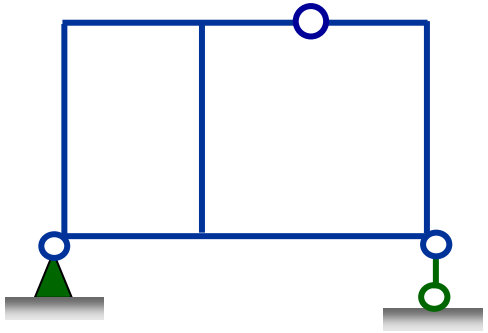
多余约束不唯一

4度

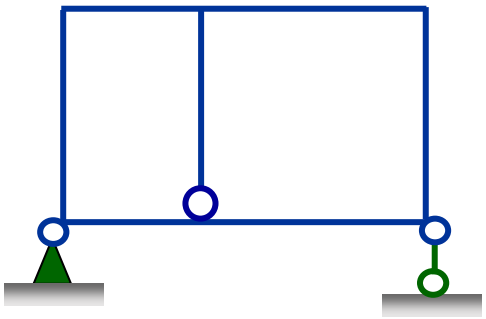




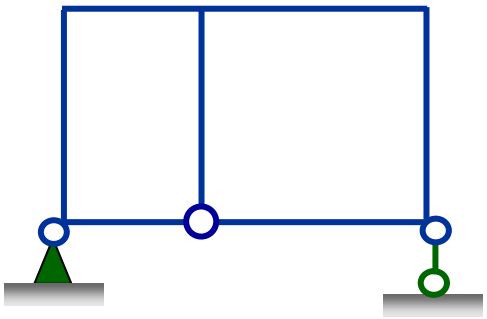
6度



5度



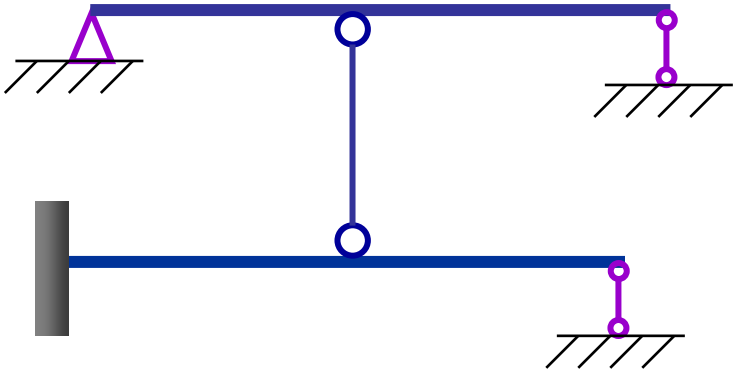
5度



4度

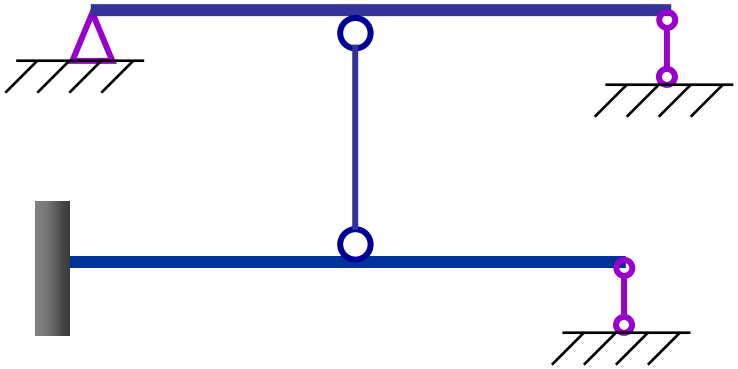
2、静不定度的判断

3）梁杆结构

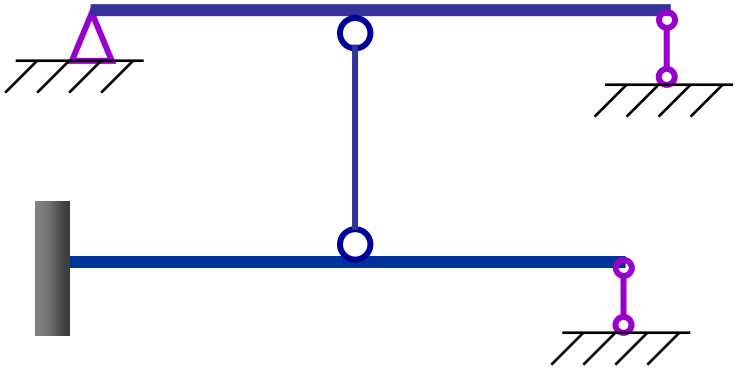


2度

安装法

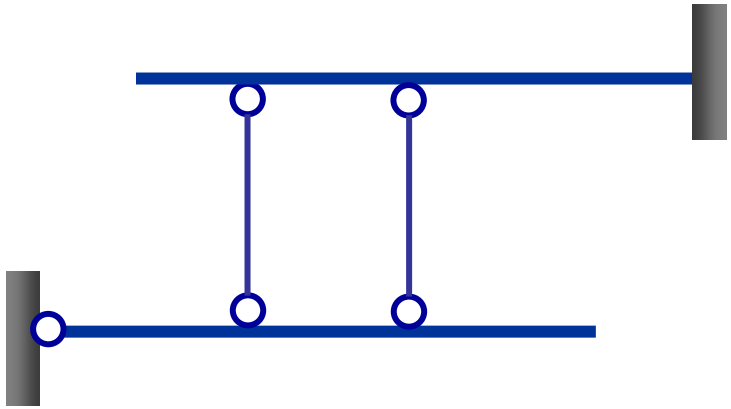


拆卸法



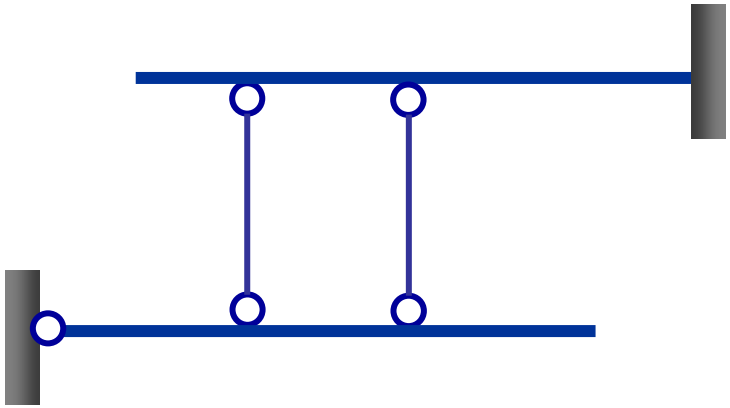
2、静不定度的判断

3）梁杆结构



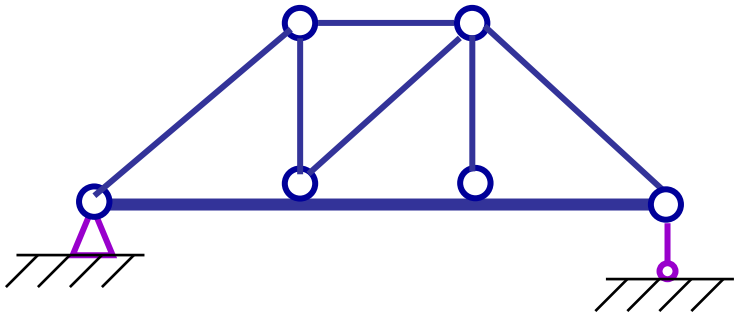
1度

拆卸法

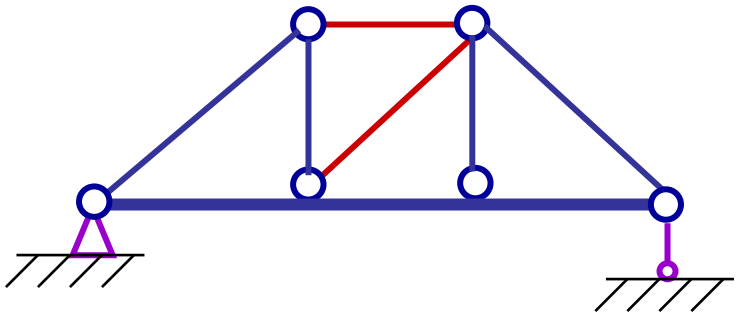


2、静不定度的判断

3）梁杆结构



2度



小结：

- 静不定度＝多余约束数
  ＝多余约束反力数
  ＝补充方程数
- 多余约束、多余约束反力不唯一

第十四章 静不定问题分析

14.2 用力法分析静不定问题

1、力法

2、外力静不定问题分析

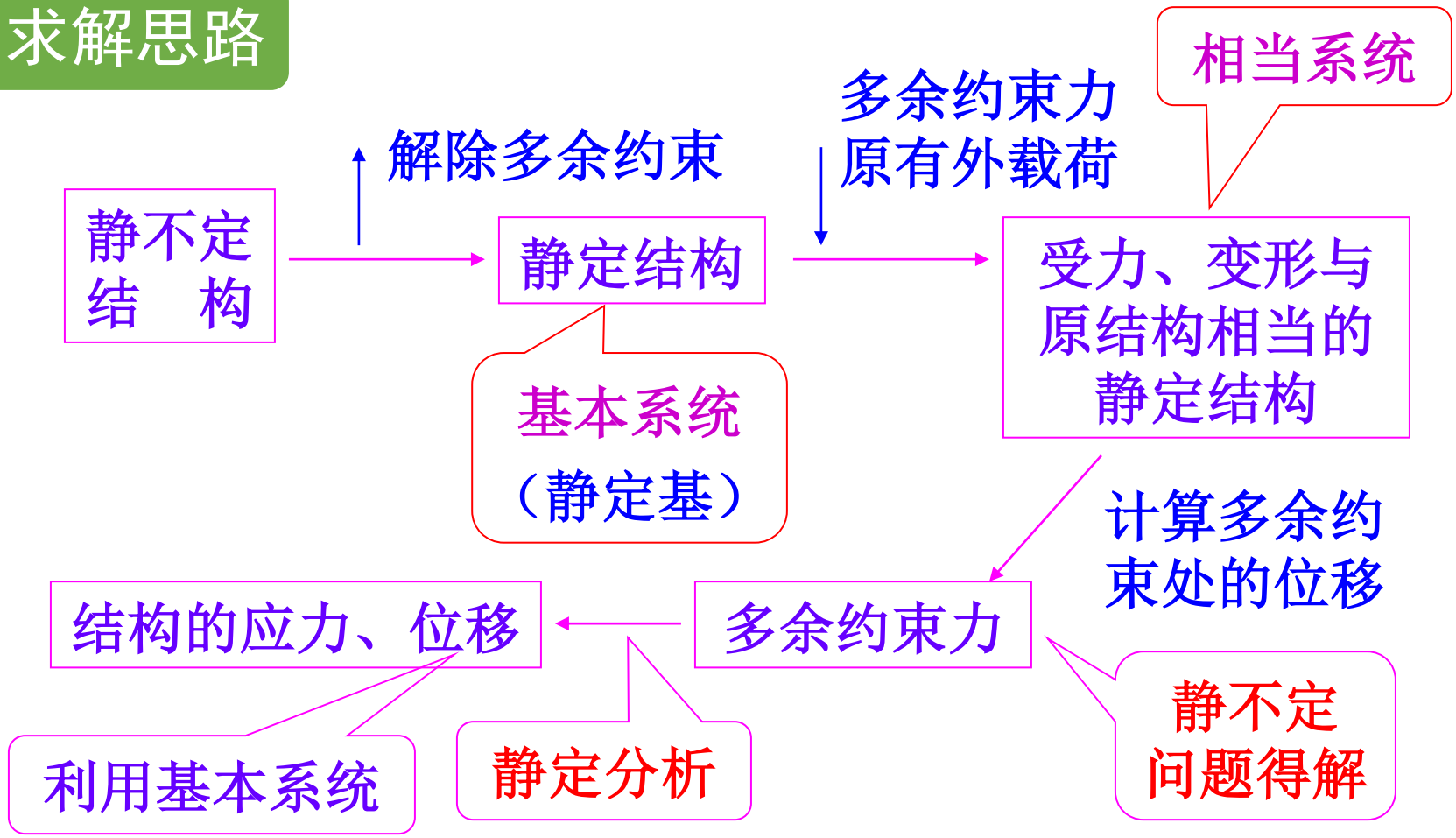
3、内力静不定问题分析

4、例题

1、力法

以多余未知力为基本未知量，求解静不定问题

求解思路



1、力法

求解步骤

1）判断静不定度，选择合适的多余约束；

2）解除多余约束，施加多余约束力，建立相当系统；

3）建立变形协调条件：

- 位移边界条件——对于外部多余约束情况；
- 位移连续条件——对于内部多余约束情况。

4）求解多余约束处的位移，由协调条件确定多余约束力；

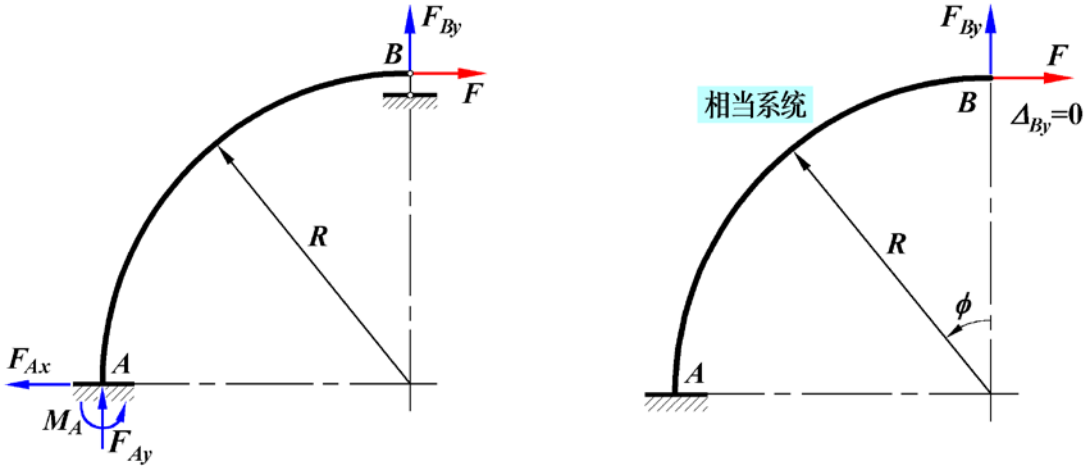
5）求静不定结构位移时，利用基本系统建立单位载荷状态

多余约束、多余约束力不唯一

基本系统、相当系统不唯一

2、外力静不定问题分析

分析图示小曲率杆的支反力与内力，*EI* 为常数。



问题转化为静定的；根据已知位移，反求未知力

1）问题分析

- 一度外力静不定；
- 选支座*B*为多余约束，建立相当系统；
- 变形协调条件（位移边界条件）

$$\boxed{\Delta_{By} = 0}$$

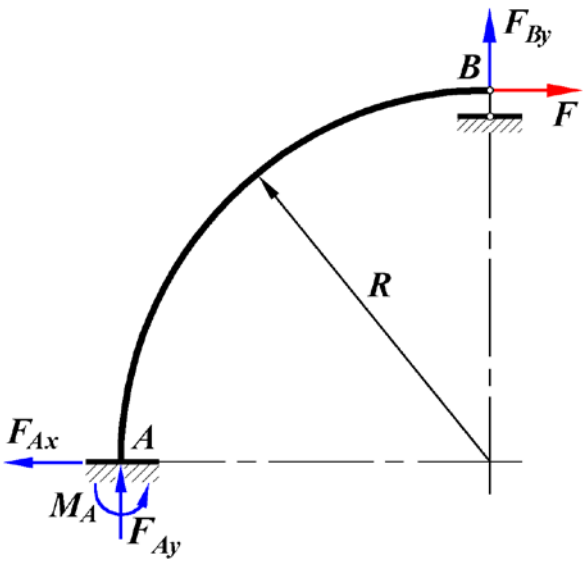
2、外力静不定问题分析

2）求位移

$$\overline{M}(\varphi) = -R\sin\varphi$$

$$M(\phi) = FR(1-\cos\phi) - F_{By}R\sin\phi$$

$$\Delta_{By} = \frac{1}{EI}\int_0^{\pi/2}\overline{M}(\varphi)M(\varphi)R\mathrm{d}\varphi = \frac{(\pi F_{By} - 2F)R^3}{EI}$$

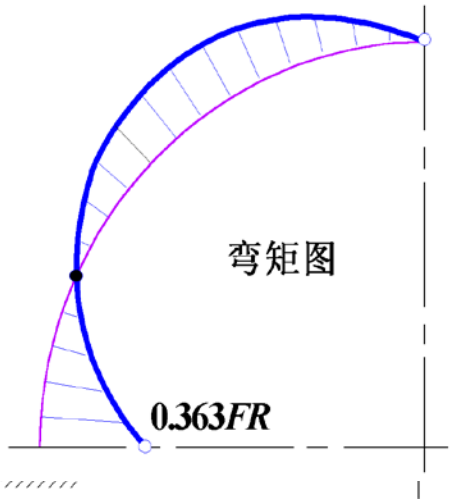


3）多余反力计算

$$\frac{(\pi F_{By} - 2F)R^3}{EI} = 0 \qquad F_{By} = \frac{2F}{\pi}$$

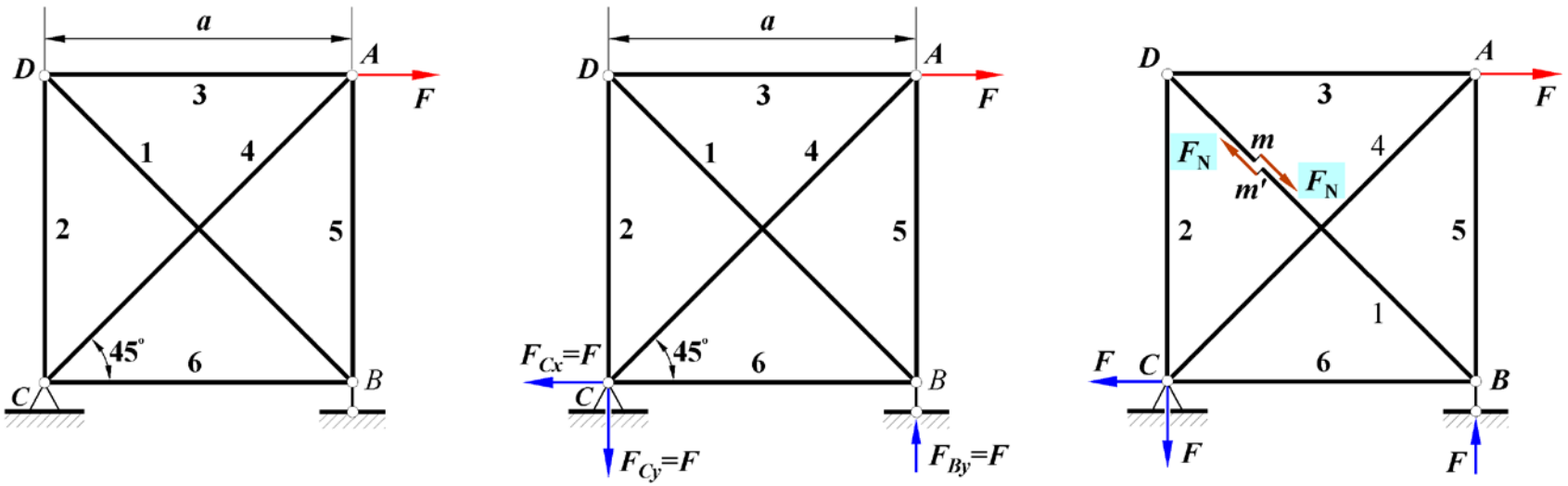
4）支反力与内力分析

$$F_{Ax} = F,\quad F_{Ay} = -\frac{2F}{\pi},\quad M_A = \left(1-\frac{2}{\pi}\right)FR$$



3、内力静不定问题分析

分析图示桁架的内力与CD杆的转角θ_{CD}，各杆EA相同



1）问题分析

- 一度内力静不定；
- 选1杆为多余约束，切开1杆，建立相当系统；
- 变形协调条件（位移连续条件）

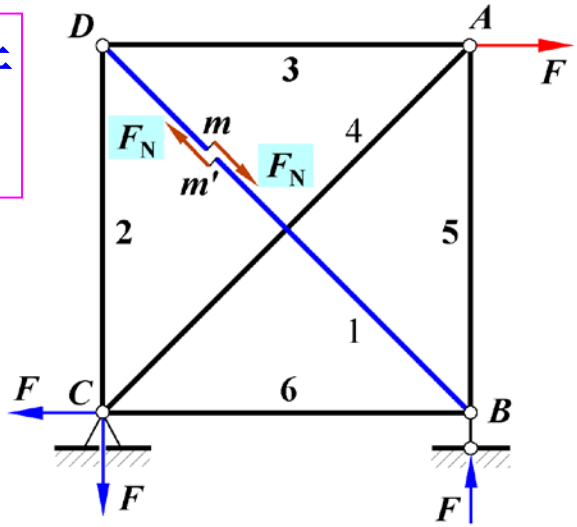
$$\Delta_{m/m'}=0$$

截面m与m'间沿轴线方向的相对线位移为零

3、内力静不定问题分析

2）加单位载荷

满足变形协调条件
确定多余反力

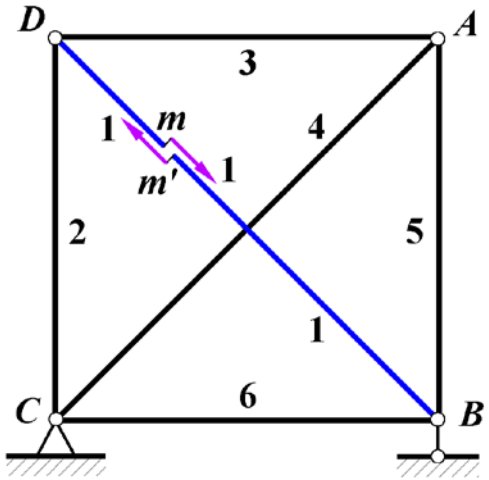


轴力列表

$$\Delta_{m/m'} = \sum_{i=1}^{6} \frac{\bar{F}_{Ni1} F_{Ni} l_i}{E_i A_i} = \frac{a}{EA}\left[\left(2+2\sqrt{2}\right)F_N + \frac{4+\sqrt{2}}{2}F\right]$$

$$\Delta_{m/m'} = 0$$

$$\Rightarrow \quad F_N = -\frac{3\sqrt{2}-2}{4}F = -0.561F$$



思考：

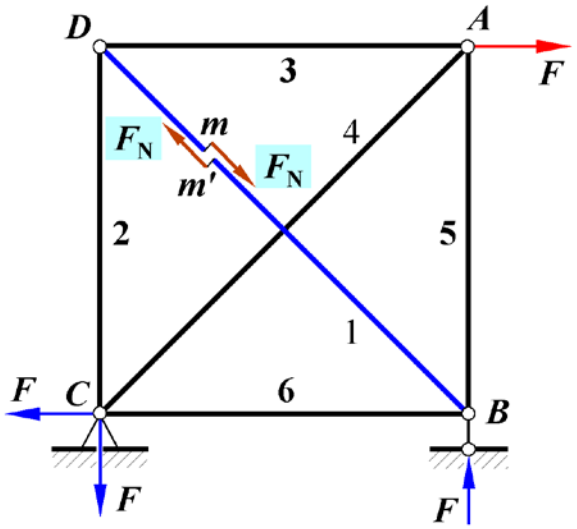
可否去掉DB杆建立相当系统？

相应的变形协调条件是什么？

3、内力静不定问题分析

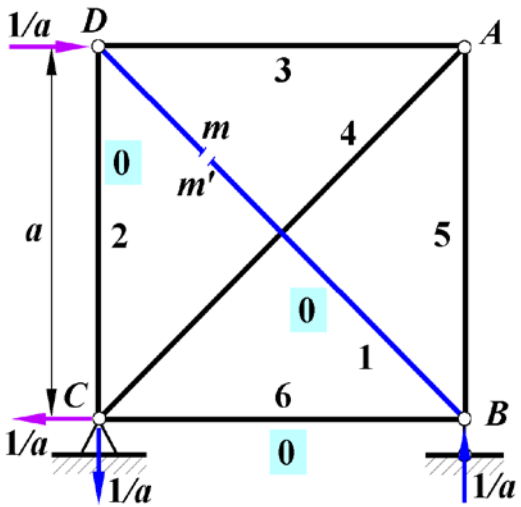
3）转角计算

$$\theta_{CD}=\sum_{i=1}^{6}\frac{\bar{F}_{Ni2}F_{Ni}l_i}{E_iA_i}$$



轴力列表

$$\theta_{CD}=\sum_{i=1}^{6}\frac{\bar{F}_{Ni2}F_{Ni}l_i}{E_iA_i}=1.912\frac{F}{EA}\quad(\circlearrowright)$$

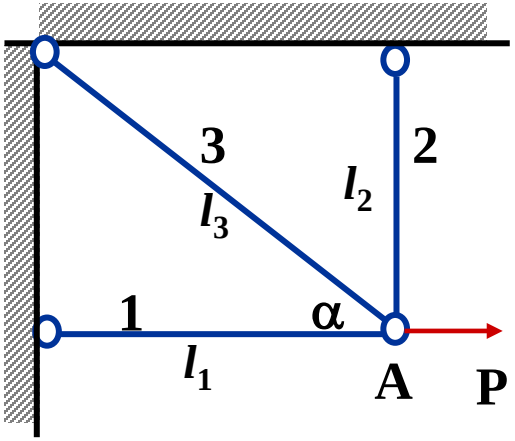


在基本系统上加单位力求位移！

为什么？

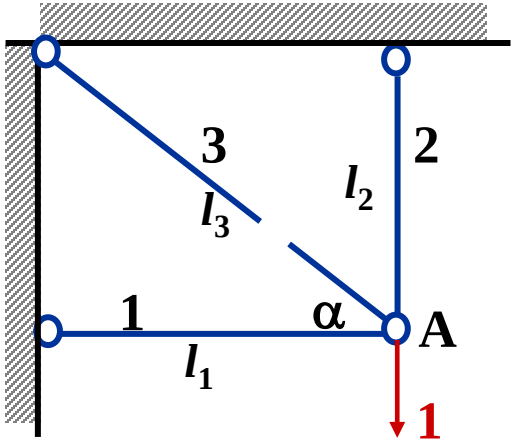
思考题

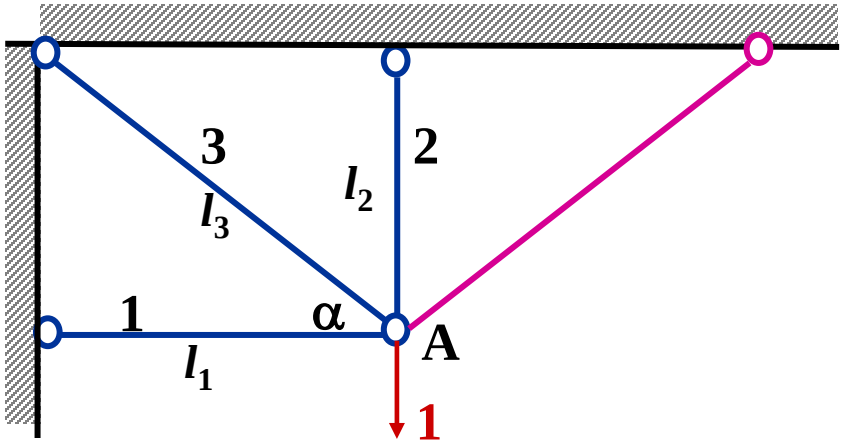
图示桁架，各杆*EA*相同，求*A*点的铅垂位移。



在求出各杆轴力后，求A点的铅垂位移：

选取单位载荷状态：

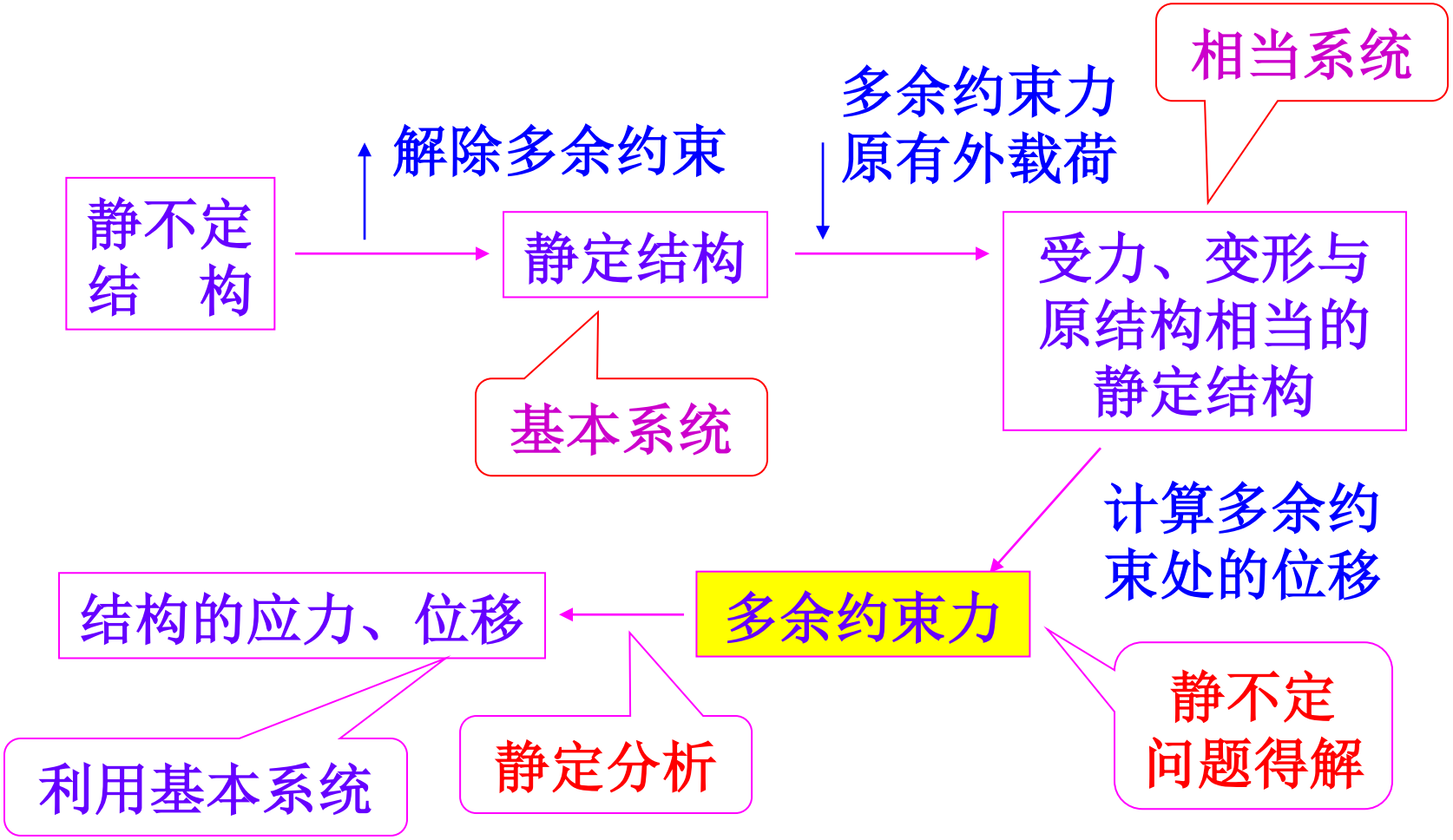




单位力加在基本系统上，为什么？

该单位载荷状态是否可用？

小结：力法求解思路


静不定结构
解除多余约束
静定结构
基本系统
多余约束力
原有外载荷
相当系统
受力、变形与原结构相当的静定结构
计算多余约束处的位移
多余约束力
静不定问题得解
静定分析
结构的应力、位移
利用基本系统

小结：力法求解思路

多余约束、多余约束力不唯一

基本系统：解除静不定结构中的多余约束后所得静定结构，也叫静定基；

相当系统：解除多余约束，代之以多余约束反力，受力、变形与原结构相当的静定结构

基本系统、相当系统不唯一

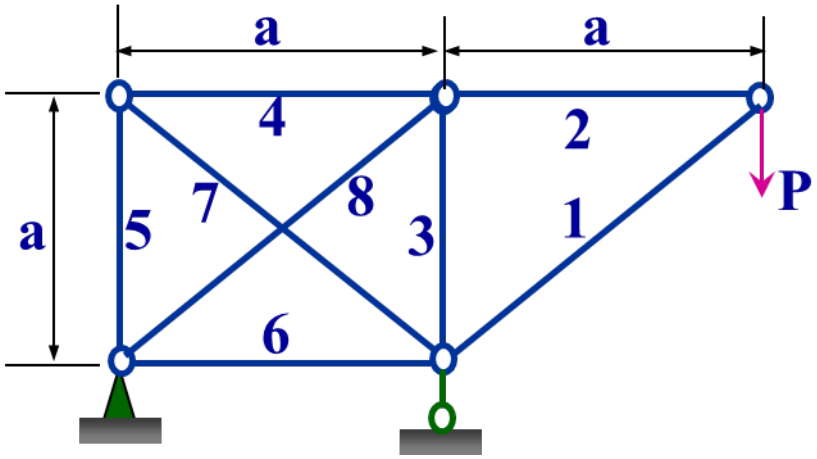
补充方程的形式

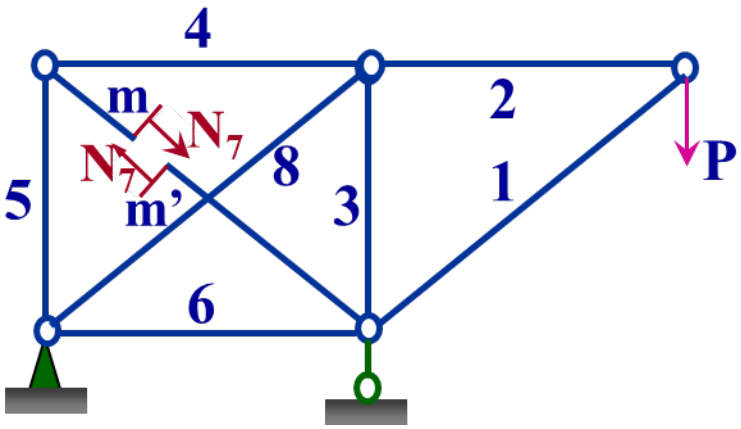
- 支座处位移边界条件 —— 外力静不定
- 结构内部位移连续条件 —— 内力静不定

在基本系统上加单位力求位移

4、例题

例14-1 图示桁架，各杆EA相同，求各杆轴力。



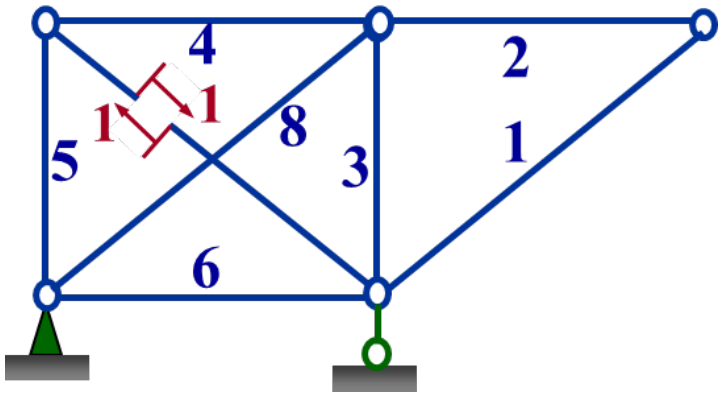


解： 1）静不定度 $8 - 2\times5 + 3 = 1$

2）切开7杆，建立相当系统

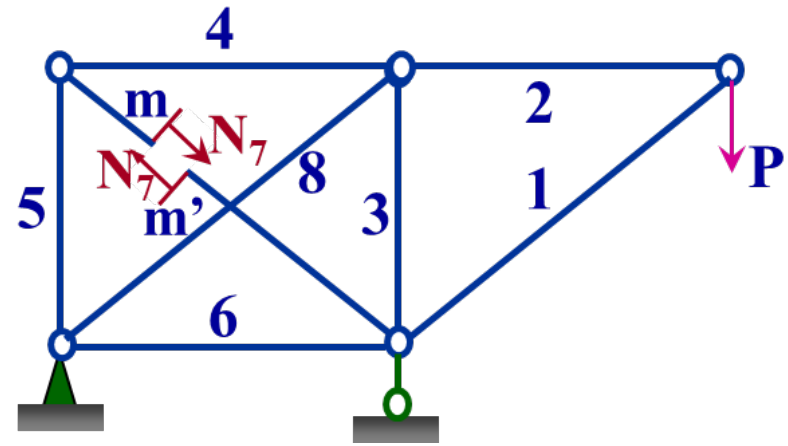
变形协调条件 $\Delta_{m/m'} = 0$

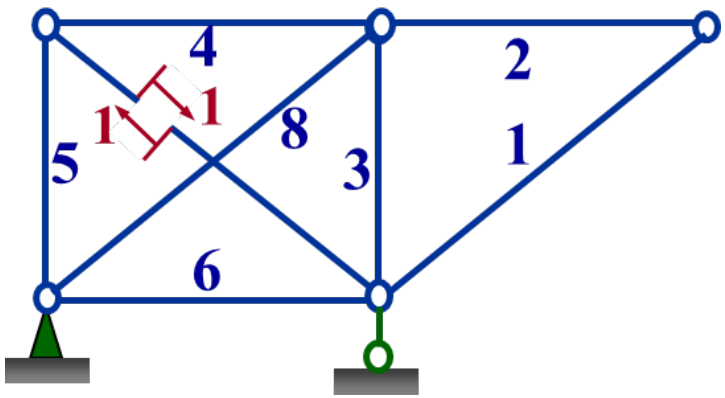
3）加单位力求相对位移



4、例题

3）加单位力求相对位移





轴力列表

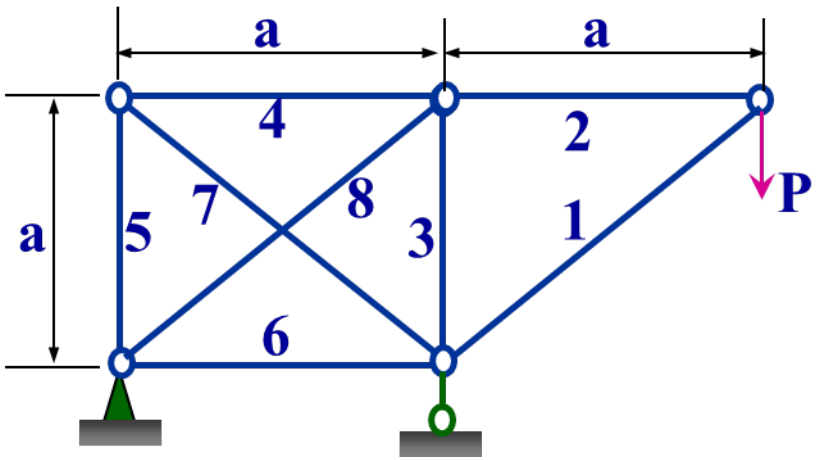
	l_i	N_i	$\bar{N}_i$
1	$\sqrt{2}a$	$-\sqrt{2}P$	0
2	a	P	0
3	a	$-\frac{\sqrt{2}}{2}N_7-P$	$-\frac{\sqrt{2}}{2}$
4	a	$-\frac{\sqrt{2}}{2}N_7$	$-\frac{\sqrt{2}}{2}$
5	a	$-\frac{\sqrt{2}}{2}N_7$	$-\frac{\sqrt{2}}{2}$
6	a	$-\frac{\sqrt{2}}{2}N_7-P$	$-\frac{\sqrt{2}}{2}$
7	$\sqrt{2}a$	N_7	1
8	$\sqrt{2}a$	$N_7+\sqrt{2}P$	1

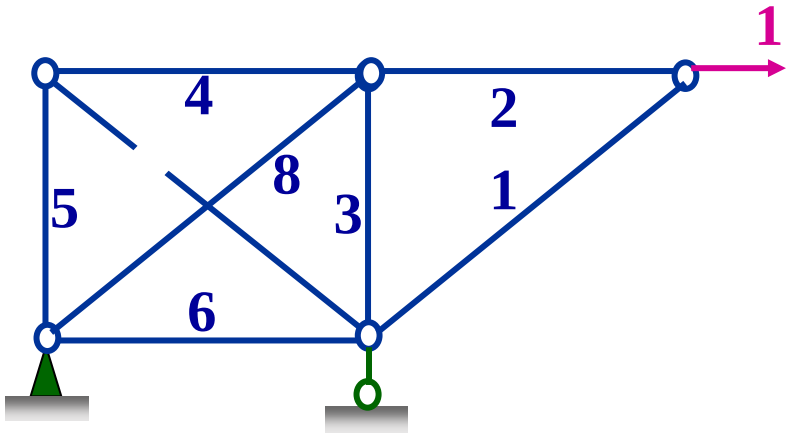
$$\Delta_{m/m'}=\sum_{i=1}^{8}\frac{\bar{N}_i N_i l_i}{EA}=0 \quad \Rightarrow \quad N_7=-\frac{\sqrt{2}}{2}P$$

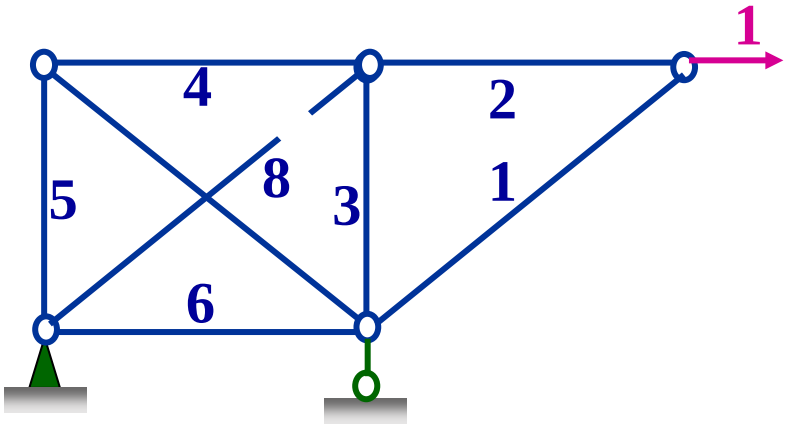
4、例题

4）若求加载点水平位移

如何选择单位载荷状态？

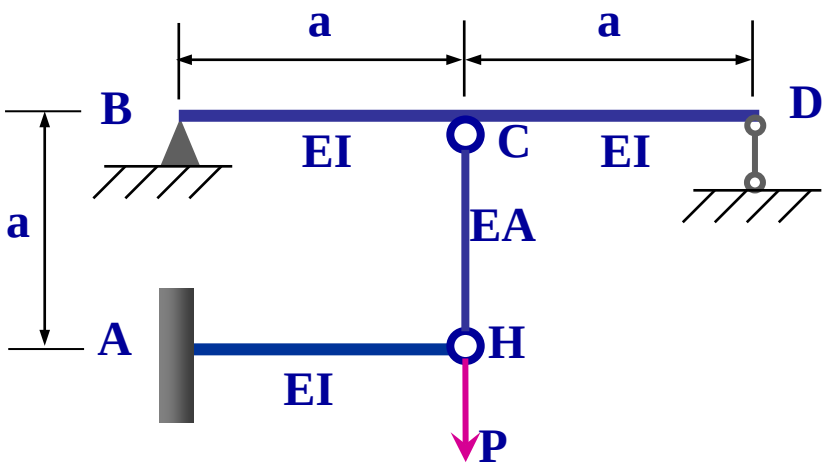






4、例题

例14-2 图示梁杆结构，已知$I=Aa^2/10$, 求杆CH的轴力以及节点H的垂直位移。



解：➢ 求CH杆的轴力：

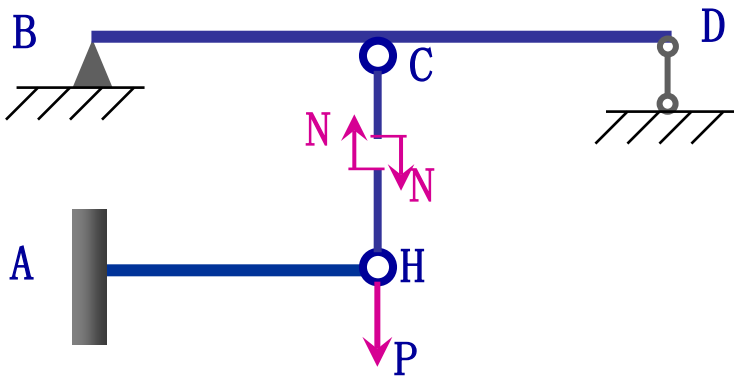
判断静不定度：

组合结构：1度内力静不定

1、 去除多余约束，建立相当系统

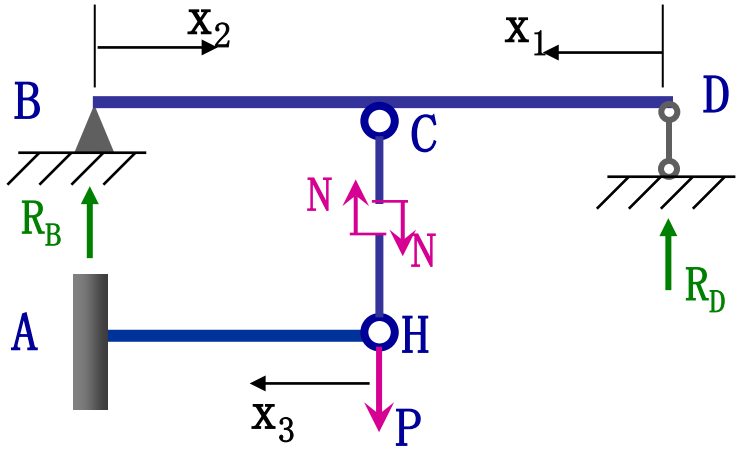
2、 建立补充方程(找变形协调条件)

$$\Delta_{m/m'}=0$$



利用单位载荷法建立补充方程

4、例题



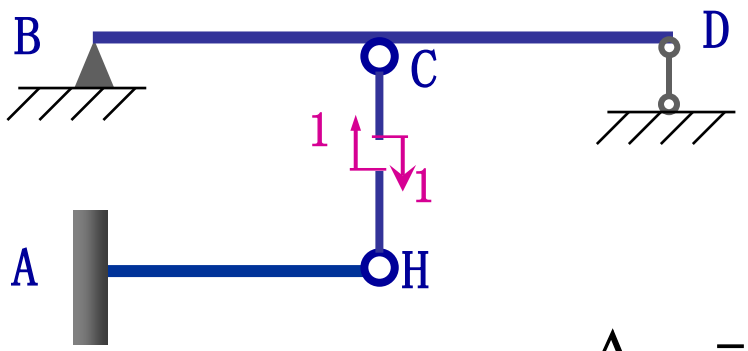
真实载荷状态(相当系统)：

$$R_B = R_D = \frac{N}{2}$$

$$M(x_1) = \frac{N}{2}x_1 \quad M(x_2) = \frac{N}{2}x_2$$

$$M(x_3) = (N - P)x_3$$

单位载荷状态：



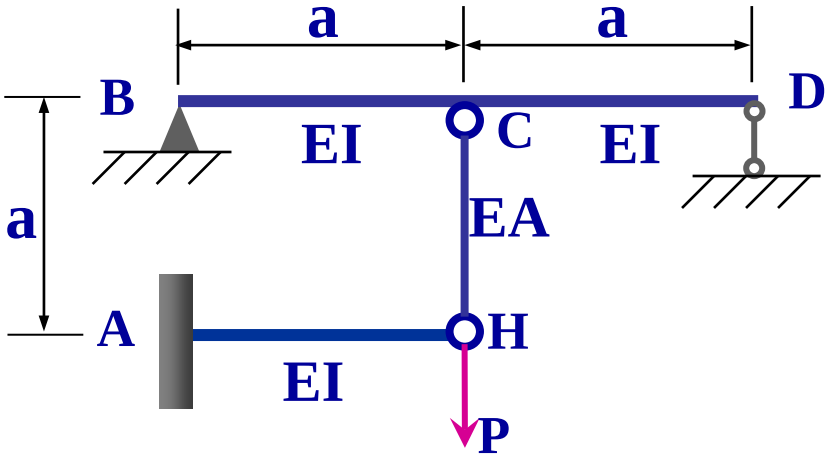
$$\bar{M}(x_1) = \frac{1}{2}x_1 \quad \bar{M}(x_2) = \frac{1}{2}x_2$$

$$\bar{M}(x_3) = x_3$$

$$\Delta_{m/m'} = \frac{1}{EI}\left(2\times\frac{N}{4}\times\frac{a^3}{3} + (N-P)\times\frac{a^3}{3}\right) + \frac{a}{EA}N$$

$$\Delta_{m/m'} = 0 \Rightarrow N = \frac{5}{9}P$$

4、例题

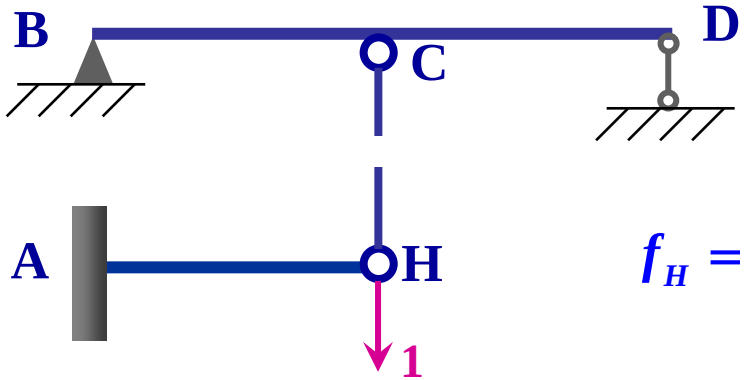


➢ 求节点H的垂直位移：

将单位载荷加在基本系统上

$$\bar{M}(x_1)=0 \quad \bar{M}(x_2)=0$$

$$M(x_3)=-x_3$$

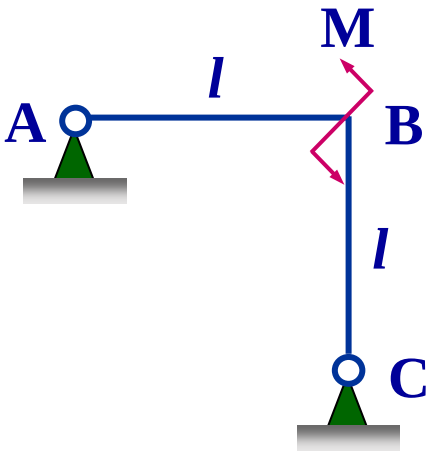


$$f_H=\frac{1}{EI}\int_0^a (N-P)x_3(-x_3)dx_3=\frac{4Pa^3}{27EI}$$

$$f_H=\frac{(P-N)a^3}{3EI}$$

4、例题

例14-3 图示刚架，已知EI为常数，求θ_A



解：

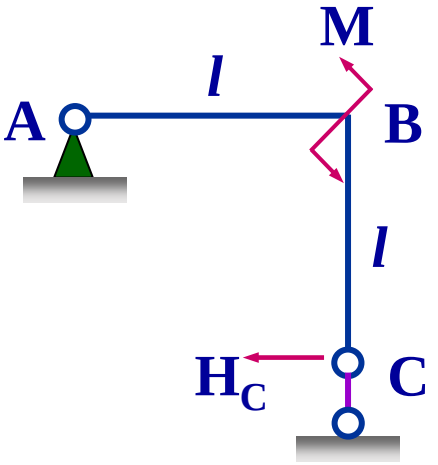
解静不定，求解多余未知力

一度外力静不定

1、 去除多余约束，建立相当系统

2、 建立补充方程(找变形协调条件)

$$\Delta_C = 0$$

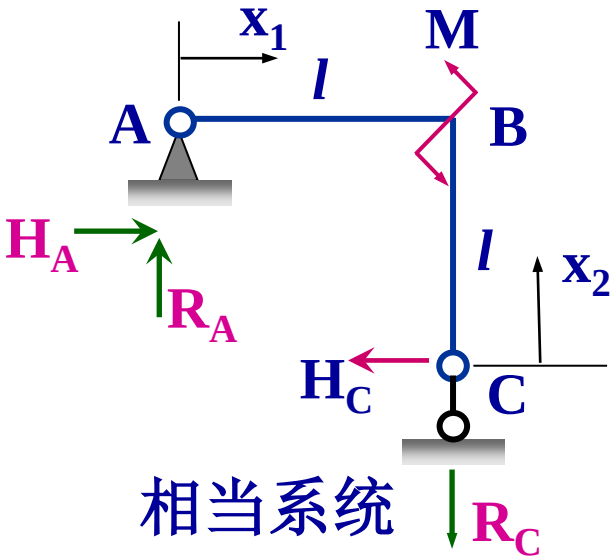


☞ 在相当系统上求Δ_C建立补充方程

可利用单位载荷法

4、例题

$$\boxed{\Delta_C = 0}$$

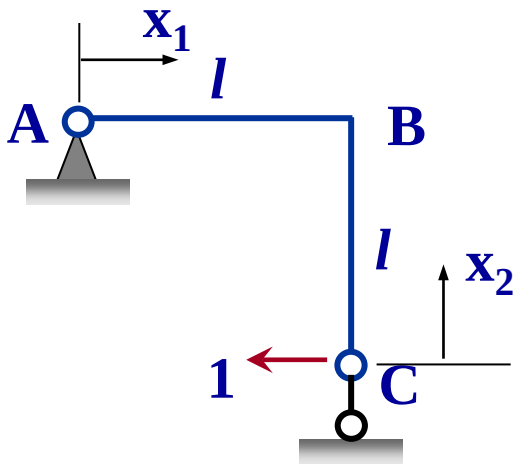


相当系统

真实载荷状态(相当系统)：

$$H_A = H_C \quad R_A = \frac{M}{l} - H_C$$

$$\boxed{\begin{aligned} M(x_1) &= (\frac{M}{l} - H_C)x_1 \\ M(x_2) &= -H_C x_2 \end{aligned}}$$



单位载荷状态

单位载荷状态：

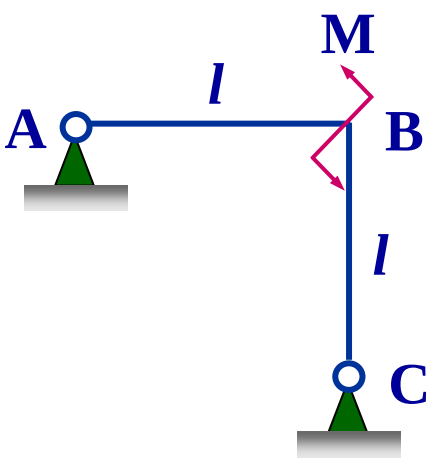
$$\boxed{\begin{aligned} \bar{M}(x_1) &= -x_1 \\ \bar{M}(x_2) &= -x_2 \end{aligned}}$$

$$\Delta_C = \frac{1}{EI}[\int_0^l \bar{M}(x_1)\cdot M(x_1)dx_1 + \int_0^l \bar{M}(x_2)\cdot M(x_2)dx_2]$$

$$\Rightarrow \quad 2H_C - \frac{M}{l} = 0 \quad \Rightarrow \quad H_C = \frac{M}{2l}$$

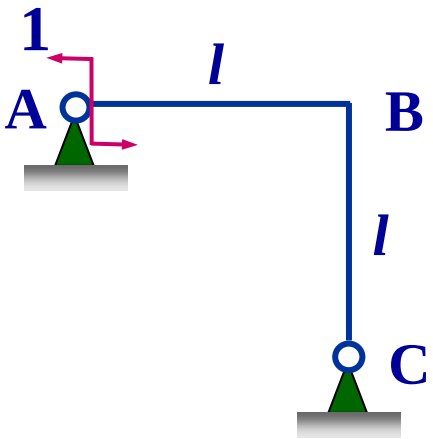
4、例题

确定H_C后，可求载荷状态的内力



以下求结构位移 θ_A

☞ 利用单位载荷法求解



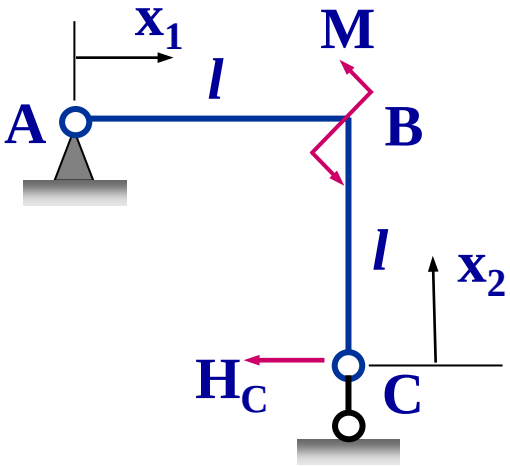
$$1\cdot\theta_A=\frac{1}{EI}[\int_0^l \bar{M}(x_1)M(x_1)dx_1+\int_0^l \bar{M}(x_2)M(x_2)dx_2]$$

可在原结构上施加单位载荷，如图

问题：还要求解静不定问题，繁琐

4、例题

将单位载荷加在基本系统上

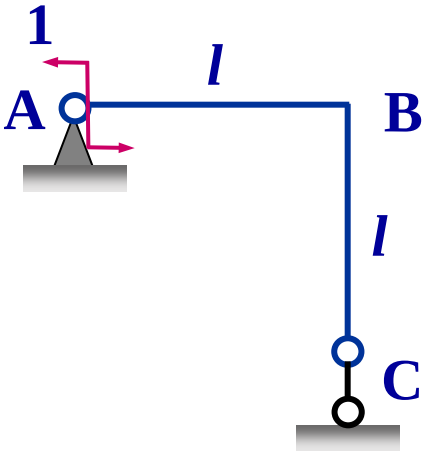


$$\boxed{\begin{aligned} M(x_1) &= \left(\frac{M}{l} - H_C\right)x_1 \\ M(x_2) &= -H_C x_2 \end{aligned}}$$

$$\bar{M}(x_1) = \frac{1}{l}x_1 - 1$$

$$\bar{M}(x_2) = 0$$

$$H_C = \frac{M}{2l}$$



$$\theta_A = \frac{1}{EI}\int_0^l \frac{M}{2l}x_1\left(\frac{1}{l}x_1 - 1\right)dx_1 = -\frac{Ml}{12EI}$$

第十四章 静不定问题分析

14.3 对称与反对称静不定问题分析

1、对称面上受力与变形特点

2、对称静不定结构受力分析

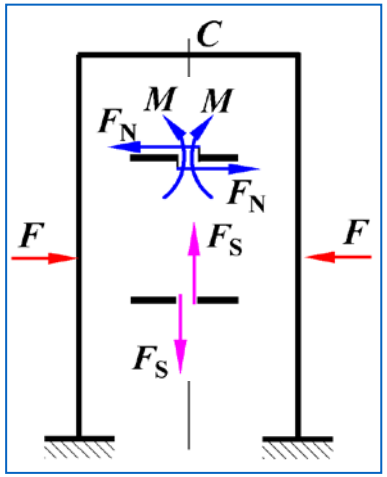
3、反对称静不定结构受力分析

4、例题

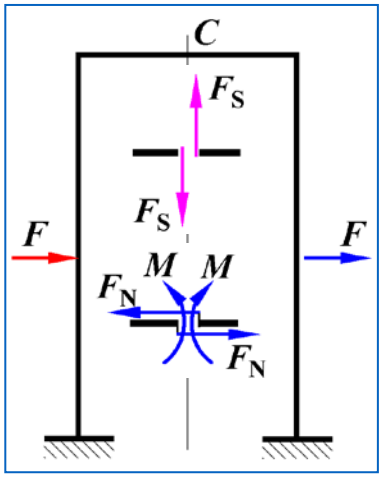
1、对称面上受力与变形特点

考察对称面两侧截面的内力与位移

对称承载



反对称承载



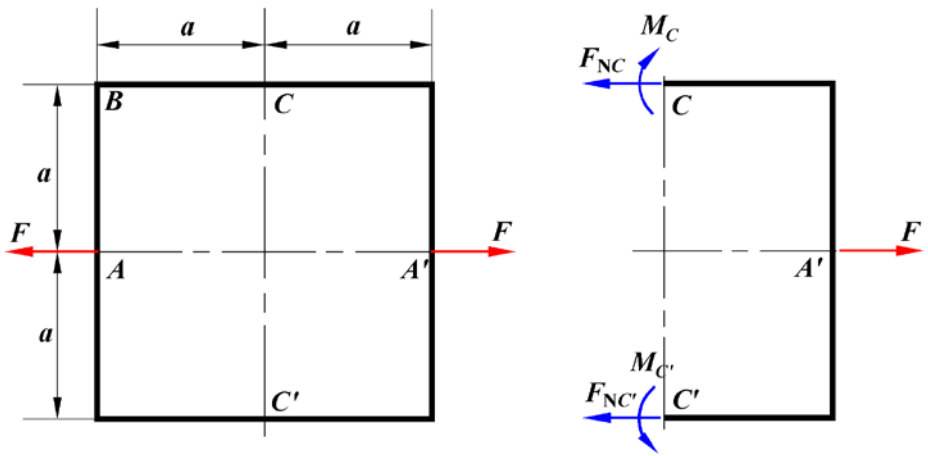
仅存在对称性内力—弯矩M与轴力F_N，
而剪力 $F_S = 0$
轴向位移 $\Delta = 0$，转角 $\theta = 0$

仅存在反对称性内力—剪力F_S
而$M=0$，$F_N = 0$
横向挠度 $f = 0$

说明：利用问题的对称性，可直接确定对称面上的多余约束力，从而简化计算分析，但并不降低静不定度

2、对称静不定结构受力分析

试分析图示刚架的弯矩，*EI*为常数



1）问题分析　3度内力静不定

存在*AA'*与*CC'*两对称轴－双对称问题

$\because$ 左右对称，$F_{S,C} = F_{S,C'} = 0$　　$\because$ 上下对称，$F_{N,C} = F_{N,C'}$

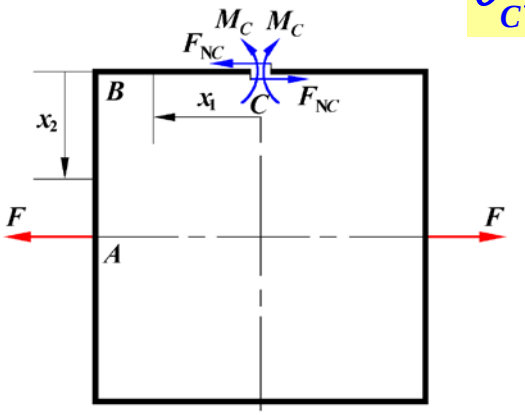
$$\sum F_x = 0,\quad F_{NC} + F_{NC'} = F \qquad F_{NC} = F_{NC'} = \frac{F}{2}$$

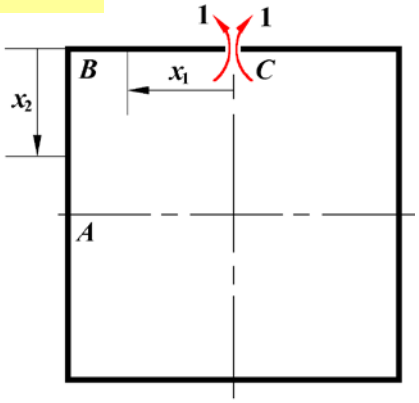
结论：一个未知多余力－M_C

2、对称静不定结构受力分析

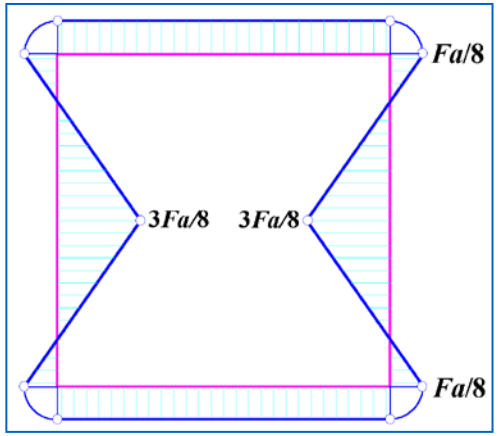
2）求解静不定

$$\theta_{C^-/C^+}=0$$





3）画弯矩图



$$|M|_{\max}=\frac{3Fa}{8}$$

4段对称

$$\theta_{C^-/C^+}=\frac{4}{EI}\left[\int_0^a \overline{M}(x_1)M(x_1)\mathrm{d}x_1+\int_0^a \overline{M}(x_2)M(x_2)\mathrm{d}x_2\right]$$

$$\overline{M}(x_1)=1,\quad \overline{M}(x_2)=1;$$

$$M(x_1)=M_C,\quad M(x_2)=M_C-\frac{Fx_2}{2}$$

$$\theta_{C^-/C^+}=\frac{(8M_C-Fa)a}{EI}=0$$

得：$M_C=\dfrac{Fa}{8}$

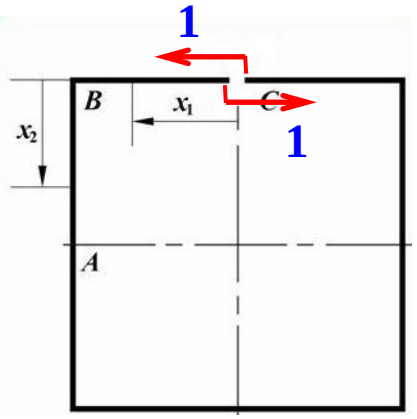
2、对称静不定结构受力分析

4）讨论：变形协调条件与单位载荷状态

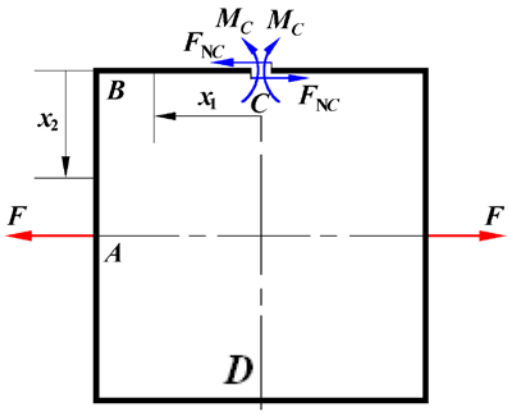
条件一：

$$\Delta_{C^-/C^+}=0$$

需求三段内力



单位载荷状态

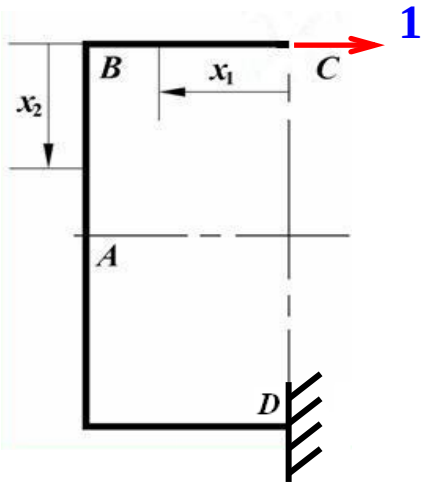


相当系统

条件二：

$$\Delta_{C/D}=0$$

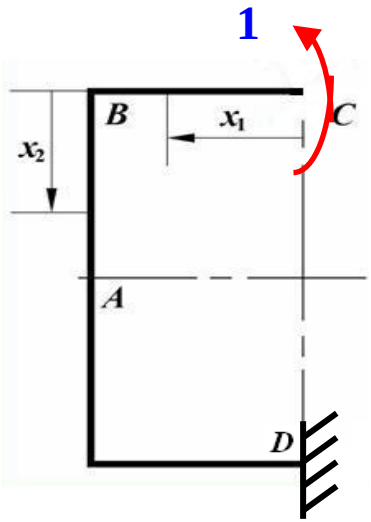
$$\Delta_C=0$$



条件三：

$$\theta_{C/D}=0$$

$$\theta_C=0$$



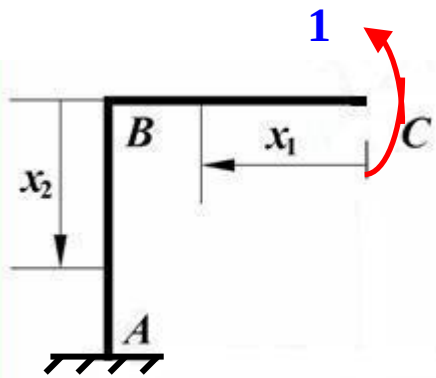
2、对称静不定结构受力分析

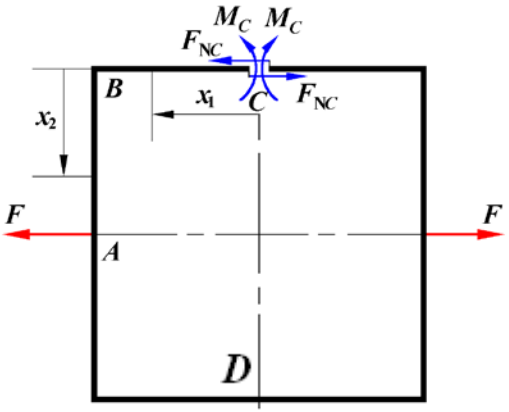
4）讨论：变形协调条件与单位载荷状态

条件四：

$$\theta_{C/A}=0$$

$$\theta_C=0$$

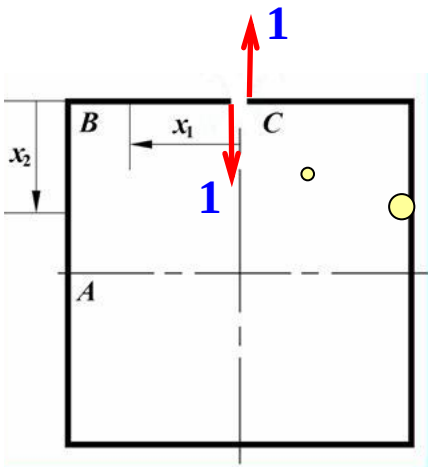




相当系统

条件五：

$$f_{C^-/C^+}=0$$



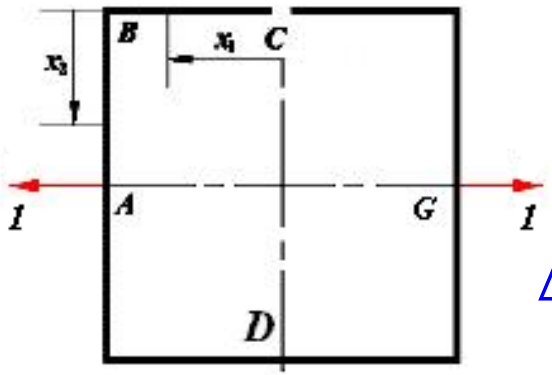
是否可用？

2、对称静不定结构受力分析

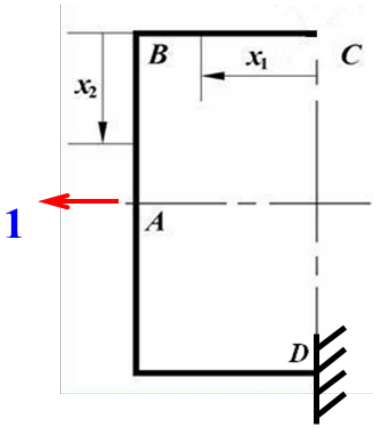
5）求载荷作用点水平方向相对位移 $\Delta_{A/G}$

如何选取单位载荷状态?

单位力加在基本系统上

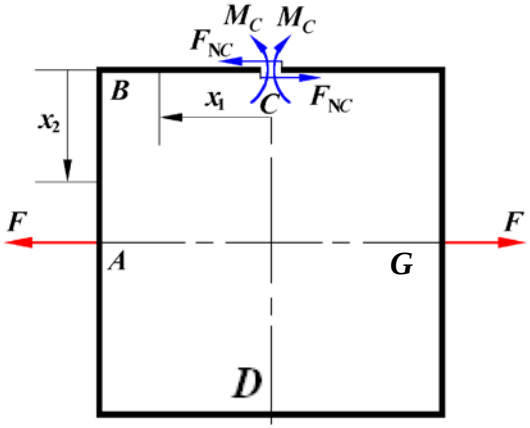


$$\Delta_{A/G}=\int_{l_{ADG}}\frac{\bar{M}_1(x)M(x)}{EI}\mathrm{d}x$$



$$\Delta_{A/D}=\int_{l_{AD}}\frac{\bar{M}_2(x)M(x)}{EI}\mathrm{d}x$$

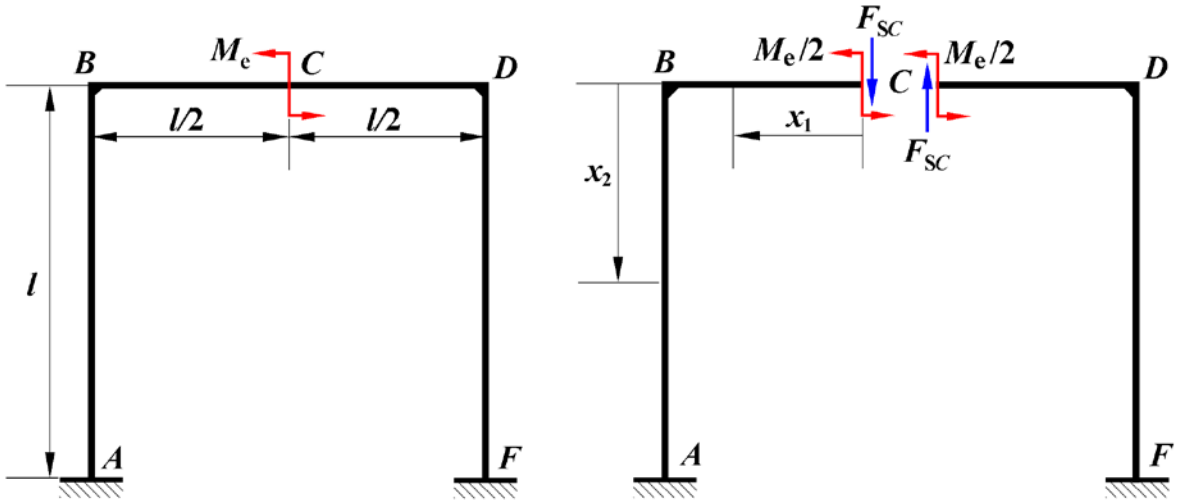
$$\Delta_{A/G}=2\Delta_{A/D}$$



相当系统

3、反对称静不定结构受力分析

试求图示刚架截面 C 的转角，EI 为常数



1）问题分析

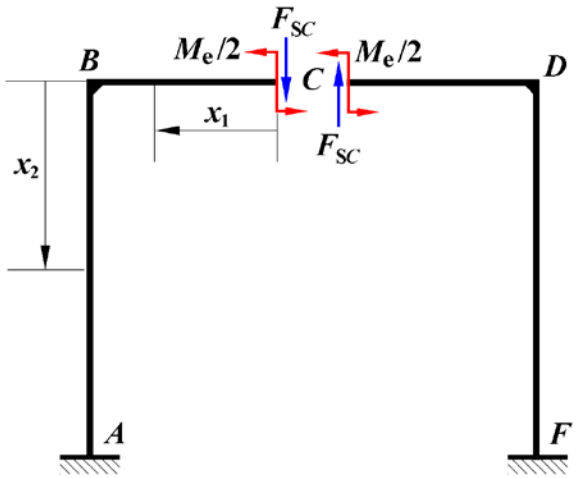
$$f_{C^-/C^+}=0$$

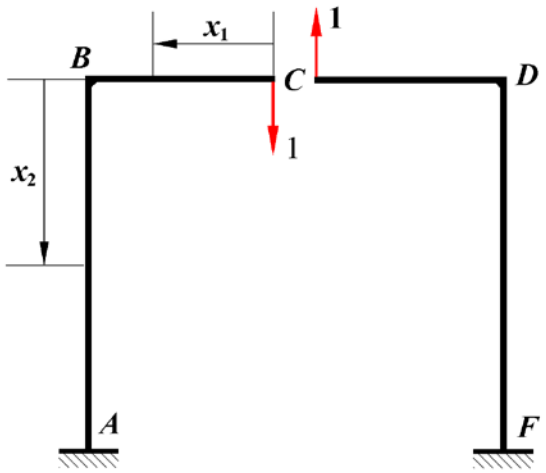
- 三度静不定；
- 结构对称，载荷反对称；
- 切开对称面，存在未知剪力F_{SC}——多余约束力；
- 建立相当系统，给出变形协调条件。

3、反对称静不定结构受力分析

2）求解多余反力——加一对力

$$\boxed{f_{C^-/C^+}=0}$$





$$M(x_1)=\frac{M_e}{2}-F_{SC}x_1,\quad M(x_2)=\frac{M_e}{2}-\frac{F_{SC}l}{2}$$

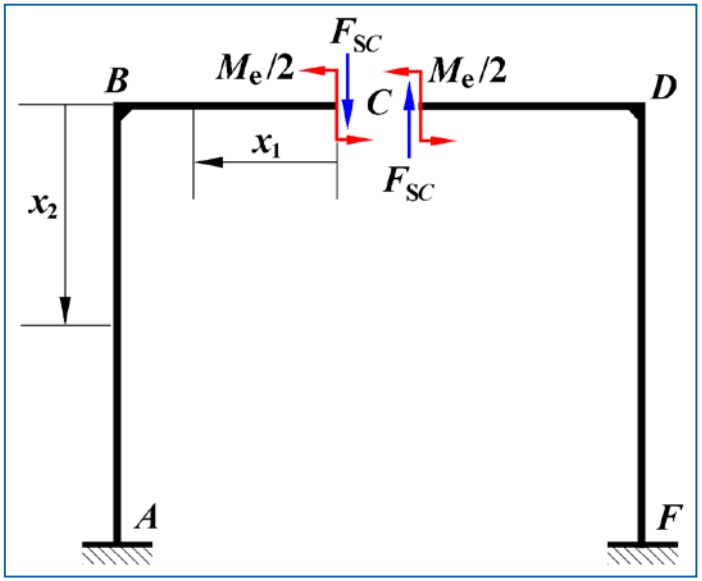
$$\overline{M}(x_1)=-x_1,\quad \overline{M}(x_2)=-\frac{l}{2}$$

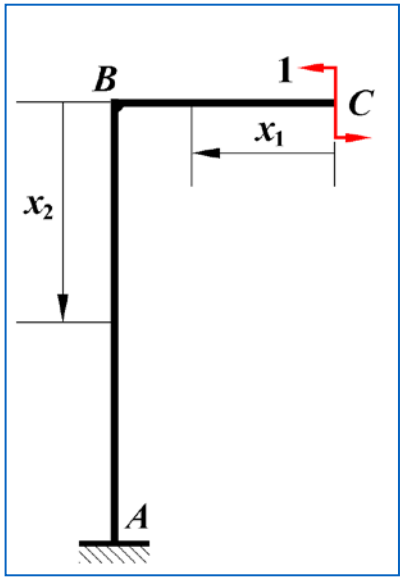
$$f_{C^-/C^+}=\frac{2}{EI}\left[\int_0^{l/2}\overline{M}(x_1)M(x_1)\mathrm{d}x_1+\int_0^{l}\overline{M}(x_2)M(x_2)\mathrm{d}x_2\right]$$

$$f_{C^-/C^+}=\frac{l^2}{24EI}\left(-15M_e+14F_{SC}l\right)=0 \quad \Rightarrow \quad \boxed{F_{SC}=\frac{15M_e}{14l}}$$

3、反对称静不定结构受力分析

3）位移计算





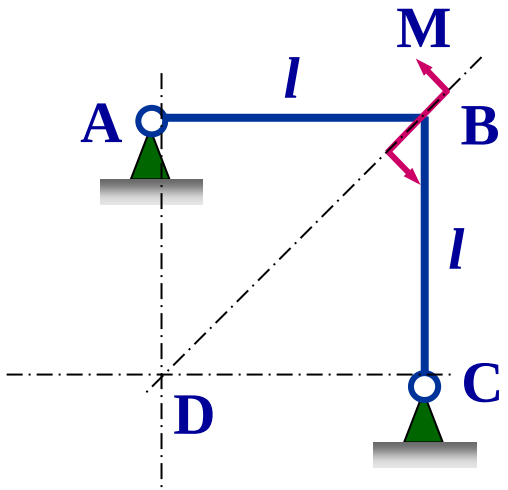
在基本系统上
加单位力求位移！

$$\theta_C = \frac{1}{EI}\left[\int_0^{l/2}\overline{M}(x_1)M(x_1)\mathrm{d}x_1 + \int_0^{l}\overline{M}(x_2)M(x_2)\mathrm{d}x_2\right]$$

$$\theta_C = \frac{9M_e l}{112EI} \quad (\circlearrowleft)$$

4、例题

例14-4 图示刚架，已知EI为常数，求支反力和θ_A



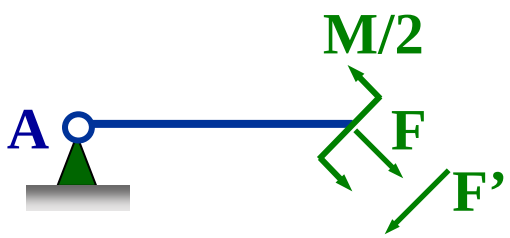
解： 一度外力静不定

结构关于BD对称，载荷反对称

沿45度方向切开B截面

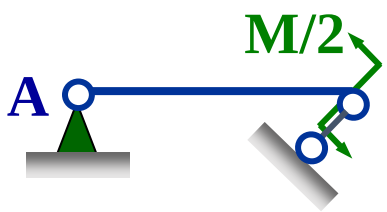
对称面处： $F=0$

$$M_B=\frac{M}{2}$$



沿BD方向的线位移为零

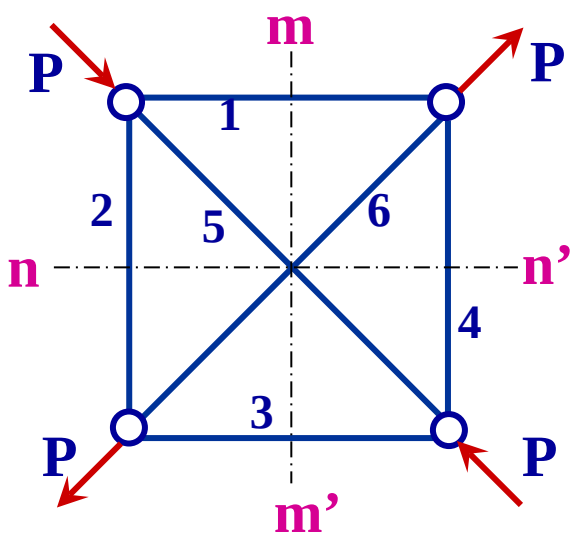
构造相当系统：



$$R_A=H_A=\frac{M}{2l}$$

4、例题

例14-5 图示桁架，已知各杆EA为常数，求轴力。



解： 结构对称，载荷反对称

对称面m-m'上：

$$N_1 = N_3 = 0$$

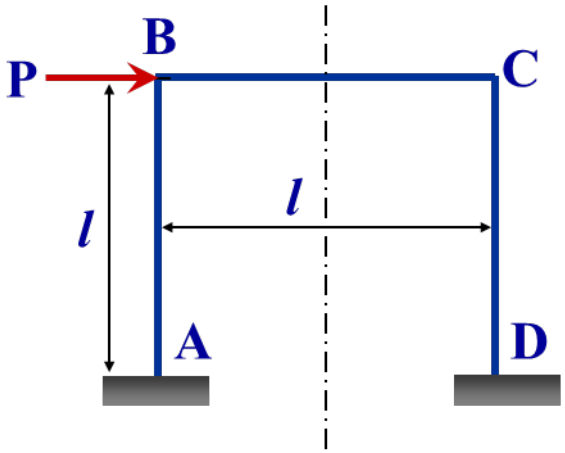
对称面n-n'上：

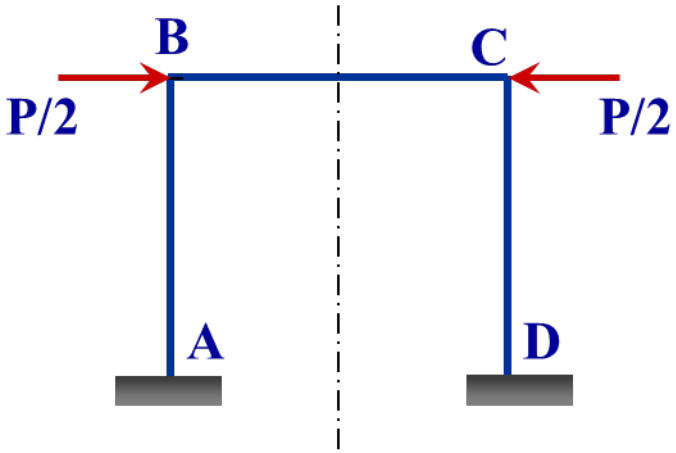
$$N_2 = N_4 = 0$$

$$\Rightarrow \quad N_5 = -P \quad N_6 = P$$

4、例题

例14-6 图示刚架，已知：P, l, EI为常数，求D端支反力.



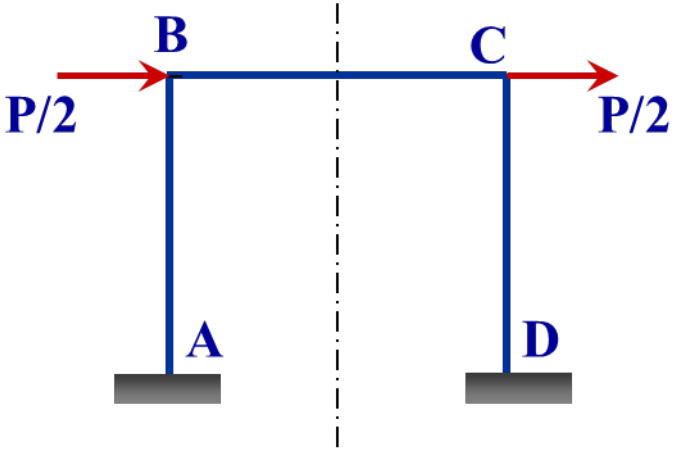


解：结构对称，载荷不对称

三度内力静不定

（三度外力静不定）

可否简化？

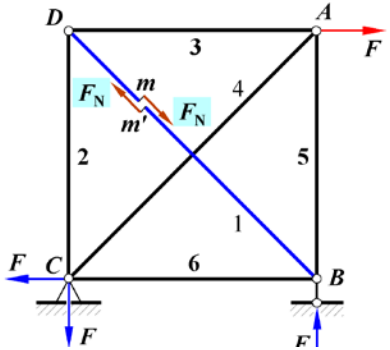


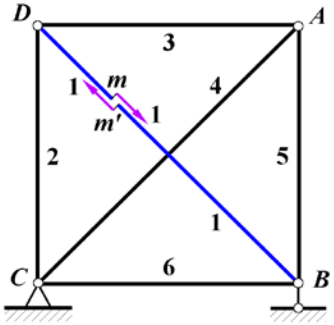
本讲：静不定结构分析

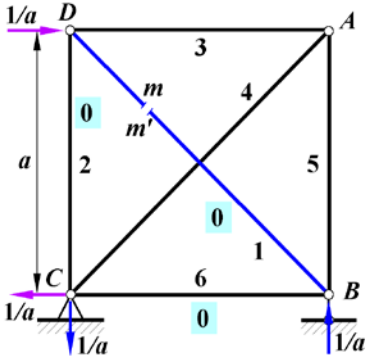
去除多余约束

下讲：平面刚架空间受力

轴力列表







	l_i	F_{Ni}	$\bar{F}_{Ni1}$	$\bar{F}_{Ni1}F_{Ni}l_i$	$\bar{F}_{Ni2}$	$\bar{F}_{Ni2}F_{Ni}l_i$
1	$\sqrt{2}a$	F_N	1	$\sqrt{2}aF_N$	0	0
2	a	$-F_N/\sqrt{2}$	$-1/\sqrt{2}$	$a/2\,F_N$	0	0
3	a	$-F_N/\sqrt{2}$	$-1/\sqrt{2}$	$a/2\,F_N$	$-1/a$	$-0.397F$
4	$\sqrt{2}a$	$\sqrt{2}F+F_N$	1	$(2F+\sqrt{2}F_N)a$	$\sqrt{2}/a$	$1.706F$
5	a	$-F-F_N/\sqrt{2}$	$-1/\sqrt{2}$	$(\sqrt{2}F+F_N)a/2$	$-1/a$	$0.603F$
6	a	$-F_N/\sqrt{2}$	$-1/\sqrt{2}$	$a/2\,F_N$	0	0
				$(2+2\sqrt{2})F_Na$ $+(2+1/\sqrt{2})Fa$		$1.912F$